

SkillSync: AI-Powered Expert Matching for Smarter Interviews

SkillSync is an innovative AI-driven solution designed to optimize interviewer selection for interview panels by ensuring alignment with candidates' areas of expertise.

Key Features:

- **Automated Profile Matching:** Uses **Sentence-BERT (SBERT)** and **Cosine Similarity** to assess relevance between experts and candidates.
- **Expert Ranking:** Generates a ranked list based on **Profile Matching Score** and **Relevancy Score** (calculated via **Weighted Sum**).
- **Secure Data Handling:** Implements **AES-256 encryption** and **OAuth 2.0 authentication** to ensure data privacy.

Technical Approach:

- **AI Algorithm:** Employs **SBERT** for text processing and **XGBoost** for ranking refinement.
- **UI/UX:** Built using **Vue.js** and **Material-UI** for an intuitive user interface.
- **Cloud & Security:** Uses **MySQL** for database management and **Azure REST APIs** for scalability.

Feasibility & Risk Mitigation:

- **Cloud-Native Architecture:** Integrates **Prometheus** and **Grafana** for auto-scaling.
- **Latency Handling:** Implements **RabbitMQ** for efficient load balancing.

- **Model Optimization:** Utilizes **distillation & pruning** to enhance SBERT efficiency.

Impact & Benefits:

- **For Candidates:** Ensures fair and relevant evaluations.
- **For Experts:** Enhances role significance and engagement.
- **For Organizations:** Speeds up hiring, reduces biases, and improves decision-making.

SkillsSync redefines recruitment by enhancing interview quality and ensuring data-driven, fair, and transparent candidate evaluation.

